Year 6 Mathematics Solutions

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Learning Strategies

Mathematics is often the most challenging subject for students. Much of the trouble comes from the fact that mathematics is about logical thinking, not memorizing rules or remembering formulas. It requires a different style of thinking than other subjects. The students who seem to be "naturally" good at math just happen to adopt the correct strategies of thinking that math requires – often they don't even realise it. We have isolated several key learning strategies used by successful maths students and have made icons to represent them. These icons are distributed throughout the book in order to remind students to adopt these necessary learning strategies:



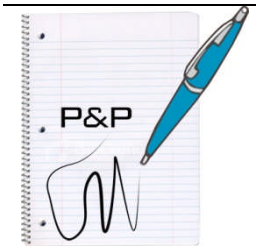
Talk Aloud Many students sit and try to do a problem in complete silence inside their heads. They think that solutions just pop into the heads of 'smart' people. You absolutely must learn to talk aloud and listen to yourself, literally to talk yourself through a problem. Successful students do this without realising. It helps to structure your thoughts while helping your tutor understand the way you think.



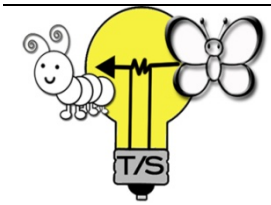
BackChecking This means that you will be doing every step of the question twice, as you work your way through the question to ensure no silly mistakes. For example with this question: $3 \times 2 - 5 \times 7$ you would do "3 times 2 is 5 ... let me check – no 3×2 is 6 ... minus 5 times 7 is minus 35 ... let me check ... minus 5×7 is minus 35. Initially, this may seem time-consuming, but once it is automatic, a great deal of time and marks will be saved.



Avoid Cosmetic Surgery Do not write over old answers since this often results in repeated mistakes or actually erasing the correct answer. When you make mistakes just put one line through the mistake rather than scribbling it out. This helps reduce silly mistakes and makes your work look cleaner and easier to backcheck.



Pen to Paper It is always wise to write things down as you work your way through a problem, in order to keep track of good ideas and to see concepts on paper instead of in your head. This makes it easier to work out the next step in the problem. Harder maths problems cannot be solved in your head alone – put your ideas on paper as soon as you have them – always!



Transfer Skills This strategy is more advanced. It is the skill of making up a simpler question and then transferring those ideas to a more complex question with which you are having difficulty.

For example if you can't remember how to do long addition because you can't recall exactly how to carry the one: $\begin{array}{r} 5889 \\ +4587 \\ \hline \end{array}$ then you may want to try adding numbers which you do know how to calculate that also involve carrying the one: $\begin{array}{r} 5 \\ +9 \\ \hline \end{array}$

This skill is particularly useful when you can't remember a basic arithmetic or algebraic rule, most of the time you should be able to work it out by creating a simpler version of the question.

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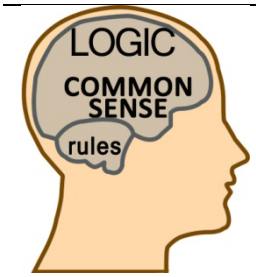
Format Skills These are the skills that keep a question together as an organized whole in terms of your working out on paper. An example of this is using the "=" sign correctly to keep a question lined up properly. In numerical calculations format skills help you to align the numbers correctly.

This skill is important because the correct working out will help you avoid careless mistakes. When your work is jumbled up all over the page it is hard for you to make sense of what belongs with what. Your "silly" mistakes would increase. Format skills also make it a lot easier for you to check over your work and to notice/correct any mistakes.

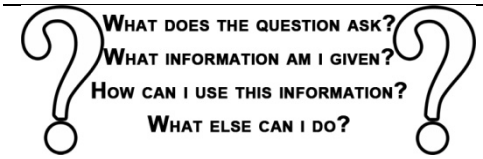
Every topic in math has a way of being written with correct formatting. You will be surprised how much smoother mathematics will be once you learn this skill. Whenever you are unsure you should always ask your tutor or teacher.



Its Ok To Be Wrong Mathematics is in many ways more of a skill than just knowledge. The main skill is problem solving and the only way this can be learned is by thinking hard and making mistakes on the way. As you gain confidence you will naturally worry less about making the mistakes and more about learning from them. Risk trying to solve problems that you are unsure of, this will improve your skill more than anything else. It's ok to be wrong – it is NOT ok to not try.



Avoid Rule Dependency Rules are secondary tools; common sense and logic are primary tools for problem solving and mathematics in general. Ultimately you must understand Why rules work the way they do. Without this you are likely to struggle with tricky problem solving and worded questions. Always rely on your logic and common sense first and on rules second, always ask Why?



Self Questioning This is what strong problem solvers do naturally when they get stuck on a problem or don't know what to do. Ask yourself these questions. They will help to jolt your thinking process; consider just one question at a time and Talk Aloud while putting Pen To Paper.

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Year 6 Mathematics

Number

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Exercise 1

Negative Numbers

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1) Complete the following patterns

a) 5, 4, 3, ____, ____, _____

2, 1, 0

b) 6, 4, 2, ____, ____

0, -2

c) 9, 6, 3, ____, ____

0, -3

d) 10, 5, ____, ____

0, -5

e) 6, 2, ____, ____

-2, -6

f) 5, 3, 1, ____, ____

-1, -3

g) 13, 7, 1, ____, ____

-5, -11

2) Place the following numbers on a number line

a) 10

b) 6

c) --1

d) 1

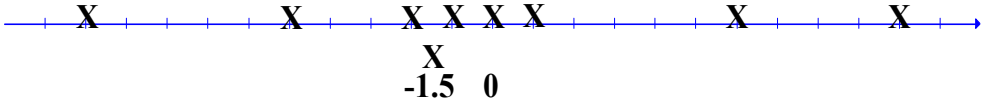
e) 0

f) –2

g) –5

h) –10

i) –1.5



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3) Calculate the following

a) $6 - 6$

0

b) $4 - 4$

0

c) $4 - 5$

-1

d) $3 - 7$

-4

e) $4 - 10$

-6

f) $0 - 3$

-3

g) $0 - 5$

-5

4) Calculate the following

a) $-2 - 3$

-5

b) $-1 - 4$

-5

c) $-5 - 2$

-7

d) $-10 - 5$

-15

e) $-3 - 12$

-15

5) Calculate the following

a) $-4 + 4$

0

b) $4 - 4$

0

c) $-2 + 4$

2

d) $4 - 2$

2

e) $-1 + 3$

2

f) $3 - 1$

2

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g) $-10 + 9$

-1

h) $9 - 10$

-1

i) $-8 + 3$

-5

j) $3 - 8$

-5

6) From your answers to question 5, rewrite the following question and hence make it easier to solve

$-a + b$

$b - a$

Where a and b are any numbers

7) The temperature at two o'clock in the afternoon was 10 degrees. Over the next 3 hours the temperature dropped by 12 degrees. What was the temperature at 5 o'clock?

$10 - 12 = -2$ Degrees

8) Canasta is a card game where at the end of each hand points are either added or deducted from a player's score. Players' scores start at zero. In the first hand Tom lost 25 points, and in the second he lost 15. What was his score after 2 hands?

$0 - 25 - 15 = -40$

9) Ian had $40 in his bank account. He wrote a cheque for $50 for petrol. Assuming his bank let the payment go through, what was the balance of Ian's account after he paid for the fuel?

$40 - 50 = -\$10$

10) In golf, every hole has a "par score" which is the amount of shots it usually takes to get the ball into the hole. Each player records how many shots it actually took, and the total score above or below "par" is recorded. The following score card shows Peter's score for 9 holes of golf.

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Hole Number	Par Score	Peter's Score	Amount above or below par	Running Total
1	5	4	--1	--1
2	4	3	--1	--2
3	4	5	+1	--1
4	3	2	-1	-2
5	5	5	0	-2
6	4	3	-1	-3
7	4	3	-1	-4
8	3	5	+2	-2
9	5	4	-1	-3

Complete the table

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Exercise 2

Place Value

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1) Write the following in numerals

a) Ten thousand three hundred and twenty seven

10327

b) Forty two thousand two hundred and twelve

42212

c) Seven thousand and seven

7007

d) Sixteen thousand and fifteen

16015

e) One hundred and twelve thousand four hundred and twenty

112420

f) One hundred and two thousand and eleven

102011

2) Write the following in words

a) 32033

Thirty two thousand and thirty three

b) 410002

Four hundred and ten thousand and two

c) 11706

Eleven thousand seven hundred and six

d) 105007

One hundred and five thousand and seven

e) 310207

Three hundred and ten thousand two hundred and seven

f) 190001

Nineteen thousand and one

3) What is the place value of the 5 in each of the following?

a) 21005

Units

b) 151443

Fifty thousand

c) 526111

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Five hundred thousand

d) 1523123

Five hundred thousand

e) 10054

Fifty

f) 500121

Five hundred thousand

4) Write the following numbers in order, from largest to smallest

2121234, 211246, 213652, 100834, 1000999, 2221011, 2021101,

2221011, 2121234, 2021101, 1000999, 213652, 211246, 100834

5) Write the following numbers in order, from smallest to largest

104224, 14425, 21501, 115001, 100516, 101010, 110110, 151002, 105509

14425, 21501, 100516, 104224, 101010, 105509, 110110, 115001, 151002

6) There were 126244 people at a soccer match. Write this number to the nearest

a) Hundred

126200

b) Thousand

126000

c) Ten thousand

130000

7) Round the number 167532556 to the nearest:

a) Ten

167532560

b) Hundred

167532600

c) Thousand

167533000

d) Ten thousand

167530000

e) Hundred thousand

167500000

f) Million

168000000

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8) Add the following

a) 3270 + five hundred and seventy five

$3270 + 575 = 3845$

b) Two thousand and nine + 1747

$2009 + 1747 = 3756$

c) One hundred and twenty thousand one hundred + eighteen thousand two hundred and twelve

$120100 + 18212 = 138312$

d) 19143 + three thousand one hundred and two

$19143 + 3102 = 22245$

e) 17111 + ninety nine

$17111 + 99 = 17210$

9) Which numeral represents hundreds in the number 323068

The number 0

10) If 500,000 is added to the number 1,686,400, which numerals change place value?

The 4 that represents hundred thousand changes to a 1

The 1 that represents millions changes to a 2

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Exercise 3

Operations on Whole Numbers

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1) Add the following

a) 254 + 236

490

b) 177 + 427

604

c) 281 + 945

1226

d) 139 + 553

692

e) 336 + 624

960

f) 172 + 228

400

2) Subtract the following

a) $499 - 544$

-45

b) $813 - 632$

181

c) $167 - 146$

21

d) $710 - 511$

199

e) $884 - 193$

691

f) $757 - 459$

298

3) Add the following

a) 293 + 682

975

b) 164 + 461

625

c) 373 + 513

886

d) 1112 + 1030

2142

e) 2146 + 119

2265

f) 1637 + 104

1741

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4) Subtract the following

a) $2174 - 162$

2012

b) $3012 - 153$

2859

c) $4292 - 113$

4179

d) $8162 - 333$

7829

e) $7313 - 618$

6695

f) $9001 - 565$

8436

5) Multiply the following

a) 422×5

2110

b) 336×8

2688

c) 7×525

3675

d) 111×13

1443

e) 273×12

3276

f) 531×15

7965

6) Multiply the following

a) 34×272

9248

b) 52×289

15028

c) 161×22

3542

d) 503×41

20623

e) 665×37

24605

f) 711×19

13509

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7) Divide the following

a) $88 \div 9$

$9\frac{7}{9}$

b) $82 \div 7$

$11\frac{5}{7}$

c) $51 \div 6$

$8\frac{3}{6} = 8\frac{1}{2}$

d) $73 \div 12$

$6\frac{1}{12}$

e) $61 \div 4$

$15\frac{1}{4}$

f) $73 \div 8$

$9\frac{1}{8}$

8) Divide the following

a) $150 \div 10$

15

b) $2200 \div 10$

220

c) $1800 \div 20$

90

d) $2310 \div 10$

231

e) $147 \div 10$

14.7

f) $222 \div 10$

22.2

9) Divide the following

a) $1200 \div 100$

12

b) $6200 \div 100$

62

c) $3210 \div 100$

32.1

d) $2591 \div 100$

25.91

e) $1111 \div 100$

11.11

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10) Calculate the following using the correct order of operation

a) $3 \times 6 + 9$

$(3 \times 6) + 9 = 27$

b) $11 \times 7 + 3$

$(11 \times 7) + 3 = 80$

c) $10 - 8 \div 4$

$10 - (8 \div 4) = 8$

d) $16 \div 4 + 4$

$(16 \div 4) + 4 = 8$

e) $3 \times 6 \div 3$

$(3 \times 6) \div 3 = 6$

f) $10 + 100 \div 10$

$10 + (100 \div 10) = 20$

11) Determine if the following are prime or composite by analysing their factors

a) 4

Factors are 1, 2, and 4

Number is composite

b) 11

Factors are 1, 11

Number is a prime

c) 12

Factors are 1, 2, 3, 4, 6, and 12

Number is composite

d) 100

Factors are 1, 2, 4, 5, 10, 20, 25, 50, and 100

Number is composite

e) 23

Factors are 1, 23

Number is a prime

f) 45

Factors are 1, 3, 5, 9, 15, and 45

Number is composite

g) 51

Factors are 1, 3, 17, and 51

Number is composite

h) 77

Factors are 1, 7, 11, and 77

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Number is composite

i) 79

Factors are 1, 79

Number is a prime

j) 1243

Factors are 1, 11, 113, and 1243

Number is composite

12) A supermarket is having a special on milk. The usual price is $3 per carton, but the special price is 3 for $7.50. How much **per carton** are shoppers saving?

$\$7.50 \div 3 = \$2.50$ per carton, which is a saving of 50 cents per carton

13) The price of steak is $22.50 per kilogram. How much is this per 100 grams?

There are 10 lots of 100 grams in a kg. Therefore $\$22.50 \div 10 = \$2.25$ per 100 grams

14) Peter walked 8km. Dan walked twice as far plus 5 km. How far did Dan walk?

Dan walked $8km \times 2 = 16km$. $16km + 5km = 21km$

15) Which is the better buy and by how much per carton? Five cartons of eggs for $21.05, or 7 cartons of eggs for $30.10

$\$21.05 \div 5 = \$4.21$ per carton

$\$30.10 \div 7 = \$4.30$ per carton

The first buy is the best

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Exercise 4

Fractions & Percentages

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1) Add the following fractions

a) $\frac{1}{2}+\frac{3}{4}$

$$\frac{1}{2}+\frac{3}{4}=\frac{2}{4}+\frac{3}{4}=\frac{5}{4}$$

b) $\frac{3}{8}+\frac{1}{4}$

$$\frac{3}{8}+\frac{1}{4}=\frac{3}{8}+\frac{2}{8}=\frac{5}{8}$$

c) $\frac{1}{5}+\frac{3}{10}$

$$\frac{1}{5}+\frac{3}{10}=\frac{2}{10}+\frac{3}{10}=\frac{5}{10}=\frac{1}{2}$$

d) $\frac{1}{3}+\frac{5}{6}$

$$\frac{1}{3}+\frac{5}{6}=\frac{2}{6}+\frac{3}{6}=\frac{5}{6}$$

e) $\frac{3}{5}+\frac{1}{10}$

$$\frac{3}{5}+\frac{1}{10}=\frac{6}{10}+\frac{1}{10}=\frac{7}{10}$$

f) $\frac{5}{6}+\frac{1}{12}$

$$\frac{5}{6}+\frac{1}{12}=\frac{10}{12}+\frac{1}{12}=\frac{11}{12}$$

2) Subtract the following fractions

a) $\frac{1}{2}-\frac{1}{6}$

$$\frac{1}{2}-\frac{1}{6}=\frac{3}{6}-\frac{1}{6}=\frac{2}{6}=\frac{1}{3}$$

b) $\frac{4}{5}-\frac{1}{10}$

$$\frac{4}{5}-\frac{1}{10}=\frac{8}{10}-\frac{1}{10}=\frac{7}{10}$$

c) $\frac{1}{2}-\frac{1}{12}$

$$\frac{1}{2}-\frac{1}{12}=\frac{6}{12}-\frac{1}{12}=\frac{5}{12}$$

d) $\frac{1}{10}-\frac{3}{100}$

$$\frac{1}{10}-\frac{3}{100}=\frac{10}{100}-\frac{3}{100}=\frac{7}{100}$$

e) $\frac{3}{4}-\frac{1}{8}$

$$\frac{3}{4}-\frac{1}{8}=\frac{6}{8}-\frac{1}{8}=\frac{5}{8}$$

f) $\frac{1}{5}-\frac{11}{100}$

$$\frac{1}{5}-\frac{11}{100}=\frac{20}{100}-\frac{11}{100}=\frac{9}{100}$$

3) Calculate the following

a) $3\times\frac{1}{5}$

$$\frac{3}{1}\times\frac{1}{5}=\frac{3}{5}$$

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b) $\frac{2}{3} \times 4$

$\frac{2}{3} \times \frac{4}{1} = \frac{8}{3}$

c) $\frac{5}{12} \times 2$

$\frac{5}{12} \times \frac{2}{1} = \frac{10}{12} = \frac{5}{6}$

d) $5 \times \frac{3}{8}$

$\frac{5}{1} \times \frac{3}{8} = \frac{15}{8}$

e) $\frac{3}{8} \times 2$

$\frac{3}{8} \times \frac{2}{1} = \frac{6}{8} = \frac{3}{4}$

f) $\frac{3}{100} \times 11$

$\frac{3}{100} \times \frac{11}{1} = \frac{33}{100}$

g) $\frac{7}{10} \times 3$

$\frac{7}{10} \times \frac{3}{1} = \frac{21}{10}$

4) Calculate the following

a) $\frac{1}{2} \times 10$

$\frac{1}{2} \times \frac{10}{1} = \frac{10}{2} = 5$

b) $\frac{1}{3} \times 21$

$\frac{1}{3} \times \frac{21}{1} = \frac{21}{3} = 7$

c) $20 \times \frac{1}{4}$

$\frac{20}{1} \times \frac{1}{4} = \frac{20}{4} = 5$

d) $35 \times \frac{1}{5}$

$\frac{35}{1} \times \frac{1}{5} = \frac{35}{5} = 7$

e) $\frac{1}{6} \times 42$

$\frac{1}{6} \times \frac{42}{1} = \frac{42}{6} = 7$

f) $\frac{1}{8} \times 72$

$\frac{1}{8} \times \frac{72}{1} = \frac{72}{8} = 9$

g) $130 \times \frac{1}{10}$

$\frac{130}{1} \times \frac{1}{10} = \frac{130}{10} = 13$

h) $\frac{1}{100} \times 130$

$\frac{1}{100} \times \frac{130}{1} = \frac{130}{100} = \frac{13}{10}$

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5) Write the following as percentages

a) $\frac{1}{2}$

$\frac{1}{2} \times 100\% = 50\%$

b) $\frac{1}{5}$

$\frac{1}{5} \times 100\% = 20\%$

c) $\frac{1}{4}$

$\frac{1}{4} \times 100\% = 25\%$

d) $\frac{1}{10}$

$\frac{1}{10} \times 100\% = 10\%$

e) $\frac{1}{8}$

$\frac{1}{8} \times 100\% = 12.5\%$

f) $\frac{1}{100}$

$\frac{1}{10} \times 100\% = 10\%$

6) Calculate the following

a) $10\% \; of \; 120$

$\frac{10}{100} \times 120 = \frac{1200}{100} = 12$

b) $25\% \; of \; 200$

$\frac{25}{100} \times 200 = \frac{5000}{100} = 50$

c) $50\% \; of \; 360$

$\frac{50}{100} \times 360 = \frac{18000}{100} = 180$

d) $20\% \; of \; 120$

$\frac{20}{100} \times 120 = \frac{2400}{100} = 24$

e) $50\% \; of \; 10\% \; of \; 100$

$\frac{50}{100} \times \frac{10}{100} \times 100 = \frac{50000}{10000} = 5$

f) $25\% \; of \; 20\% \; of \; 200$

$\frac{25}{100} \times \frac{20}{100} \times 200 = \frac{50000}{10000} = 10$

7) John eats one-third of a cake and Peter eats one-sixth. What fraction of the cake has been eaten?

$\frac{1}{3} + \frac{1}{6} = \frac{2}{6} + \frac{1}{6} = \frac{3}{6} = \frac{1}{2}$

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8) On day one, Martin read one quarter of a book. On the second day he read three eighths of the book. On the third day he finished the book. What fraction of the book did he read on the third day?

$\frac{1}{4}+\frac{3}{8}=\frac{2}{8}+\frac{3}{8}=\frac{5}{8}$ of the book was read by the end of day 2

$1-\frac{5}{8}=\frac{8}{8}-\frac{5}{8}=\frac{3}{8}$ was left to read on the last day

9) Bill bought one fifth of all the apples left on a shelf. If he bought 8 apples, how many were left on the shelf?

$\frac{1}{5}\times total=8$, so total $=40$

10) John ate 25% of a cake and Bill ate one quarter. Who ate more?

$$25\%=\frac{25}{100}=\frac{1}{4}$$

They each ate the same amount

11) At a sale Mary got 50% off the price of a dress that originally cost \$40. Sally got 10% off a dress that originally cost \$300

a) Who saved most money?

$50\%\times\$40=\frac{50}{100}\times\frac{40}{1}=\frac{2000}{100}=\$20$. Mary saved \$20

$10\%\times\$300=\frac{10}{100}\times\frac{300}{1}=\frac{3000}{100}=\$30$. Sally saved \$30

Sally saved \$10 more than Mary

b) Who spent the most and by how much?

Mary spent $\$40-\$20=\$20$

Sally spent $\$300-\$30=\$270$

Sally spent \$250 more than Mary

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12) In a mathematics test Tom got $\frac{1}{4}$ of the questions wrong, and Alan got $\frac{1}{3}$ of the questions wrong. If there were 60 questions on the test who got more questions correct and by how many?

Tom got $\frac{1}{4} \times 60 = \frac{60}{4} = 15$ questions wrong. He got $60 - 15 = 45$ questions right

Alan got $\frac{1}{3} \times 60 = 20$ questions wrong. He got $60 - 20 = 40$ questions right

Tom got 5 more questions right than Alan

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Exercise 5

Decimals

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1) Express the following as decimals

a) $\frac{7}{1000}$

0.007

b) $\frac{3}{1000}$

0.003

c) $\frac{23}{1000}$

0.023

d) $\frac{1}{1000}$

0.01

e) $\frac{100}{1000}$

0.100 = 0.1

2) Express the following as fractions

a) 0.002

$0.002 = \frac{2}{1000} = \frac{1}{500}$

b) 0.035

$0.035 = \frac{35}{1000} = \frac{7}{200}$

c) 0.014

$0.014 = \frac{14}{1000} = \frac{7}{500}$

d) 0.001

$0.001 = \frac{1}{1000}$

e) 0.033

$0.033 = \frac{33}{1000}$

3) Add the following

a) 0.231 + 0.42

0.651

b) 0.15 + 0.162

0.312

c) 0.313 + 0.45

0.763

d) 0.071 + 0.28

0.351

e) 0.55 + 0.405

0.955

f) 0.83 + 0.03

0.86

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4) Add the following

a) 0.518 + 0.36

0.878

b) 0.75 + 0.108

0.858

c) 0.222 + 0.69

0.912

d) 0.054 + 0.87

0.924

e) 0.919 + 0.51

1.429

f) 0.86 + 0.418

1.278

5) Add the following

a) 1.421 + 2.11

3.532

b) 1.651 + 0.22

1.871

c) 2.35 + 1.219

3.569

d) 4.203 + 1.62

5.823

e) 5.171 + 3.11

8.281

f) 1.55 + 1.586

3.136

6) Subtract the following

a) 0.514 – 0.23

0.284

b) 0.86 – 0.135

0.725

c) 0.929 – 0.48

0.449

d) 0.777 – 0.616

0.161

e) 0.125 – 0.02

0.105

f) 0.215 – 0.24

-0.025

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7) Subtract the following

a) 1.41 – 0.621

0.789

b) 1.89 – 0.922

0.968

c) 2.12 – 0.463

1.657

d) 3.24 – 2.356

0.884

e) 9.57 – 7.194

2.376

f) 2.15 – 0.995

1.155

8) Calculate the following

a) $0.125 \div 5$

0.025

b) 0.14×5

0.7

c) $1.44 \div 8$

0.18

d) $3.5 \div 5$

0.7

e) 0.22×6

1.32

f) 0.15×8

1.2

g) $2.22 \div 3$

0.74

h) $1.5 \div 5$

0.3

9) Calculate the following

a) $2.34 \div 10$

0.234

b) 0.15×100

15

c) 1.25×1000

1250

d) 0.03×10

0.3

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e) $225.4 \div 100$

2.254

f) $1122.33 \div 1000$

1.112233

8) Tom had \$15.20, whilst Alan had ten times as much. How much money did Alan have?

$\$15.20 \times 10 = \$152$

9) The price of petrol is five times more than it was ten years ago. Today's price is \$1.40 per litre. How much was petrol per litre 10 years ago?

$\$1.40 \div 5 = 28$ cents per litre

10) After midnight the owner of a 24 hour store triples his prices. How much do each of the following cost after midnight?

a) A loaf of bread costs \$1.80 before midnight

$\$1.80 \times 3 = \$5.40$

b) A bottle of cool drink costs \$2.15 before midnight

$\$2.15 \times 3 = \$6.45$

c) A burger costs \$3.10 before midnight

$\$3.10 \times 3 = \$9.30$

d) A chocolate bar costs 95 cents before midnight

$0.95 \times 3 = \$2.85$

11) A hot dog costs \$4.65 after midnight in the store from question 10. How much does it cost before midnight?

$\$4.65 \div 3 = \$1.55$

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Year 6 Mathematics

Chance & Data

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Exercise 1

Assigning Probabilities

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1) For each of the following events

- State as a fraction the chance of the named event occurring
- Express this fraction as the simplest equivalent fraction
- Express the chance as a percentage
- (e.g. $\frac{3}{6} = \frac{1}{2} = 50\%$)

a) You spin white on a spinner with five white and five black sections

$$\frac{5}{10} = \frac{1}{2} = 50\%$$

b) You pull a black shirt out of a draw containing 2 black and 6 white shirts

$$\frac{2}{8} = \frac{1}{4} = 25\%$$

c) You roll a prime number on a six sided dice

$\frac{3}{6} = \frac{1}{2} = 50\%$. Note: 1 is not a prime number

d) A bag contains 25 balls numbered from 1 to 25; you pull out a ball whose number is divisible by 5

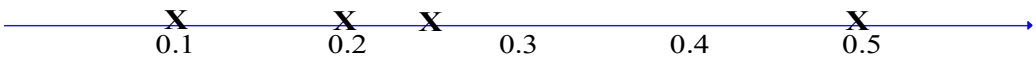
$$\frac{5}{25} = \frac{1}{5} = 20\%$$

e) A bag contains balls numbered from 1 to 100. You pull out a ball that has a number more than 19 and less than 30 on it.

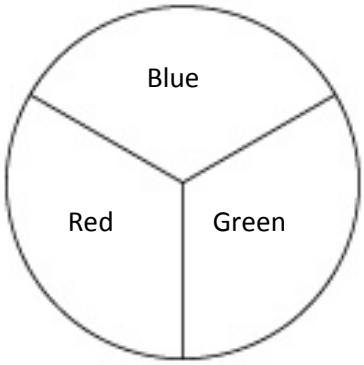
$$\frac{10}{1000} = \frac{1}{10} = 10\%$$

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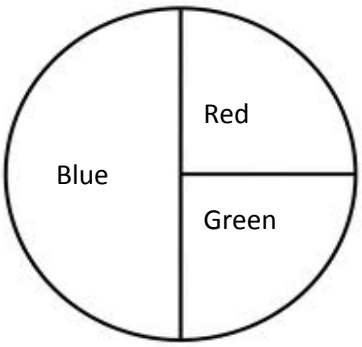
2) Place the probabilities from question 1 on a number line



3) Draw a spinner that has three colours; red, blue and green; and each colour has an equal probability of being spun

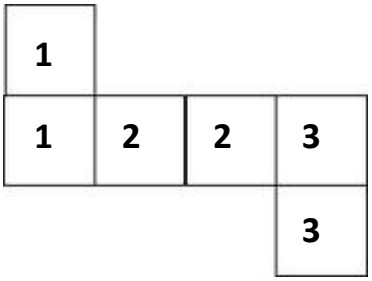


4) Draw a spinner that has three colours; red, green and blue; so that blue has twice as much chance of being spun as red or green.

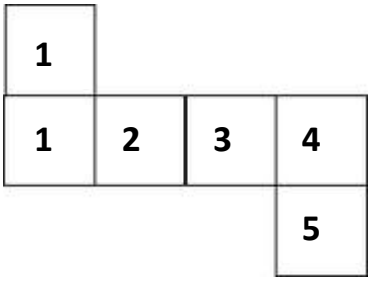


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5) Draw the six faces of a dice for which there is an equal chance of rolling the number 1, 2, or 3, but no chance of rolling any other number



6) Draw the six faces of a dice for which there is more chance of rolling a one than 2, 3, 4, or 5



7) There are 200 cubes in a bag, some white and some black. You draw out 100 of them and notice there are 52 black and 48 white. What could you say about the numbers of black and white cubes left in the bag

There are probably close to equal numbers of each colour left

8) You have to put ten shirts into a draw, some red some white and some black. How many black shirts do you have to put in the draw to make sure you have a better chance of pulling one out than any of the other colours?

Depending on the quantity of the other colours as shown in the table

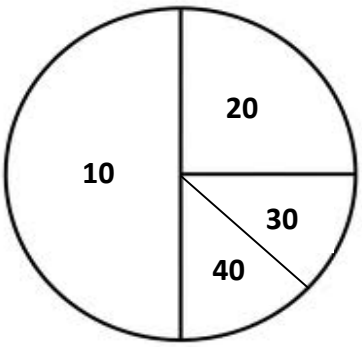
Colour	Black	Maximum of one other colour (e.g. Red)	Minimum of other colour (e.g. White)
	1	8	1
	2	7	1

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	3	6	1
	4	5	1
	5	4	1
	6	3	1
	7	2	1

Therefore if you put at least 5 black shirts in the draw you have a better than even chance of pulling a black shirt out, since the maximum of either of the other two colours is 4

9) Design a dartboard with 4 numbers on it so that you have a one in two chance of hitting the number 10, a one in four chance of hitting the number 20, and a one in eight chance of hitting the numbers 30 or 40



10) Draw a map of an imaginary world. If you close your eyes and point to a spot on your map there will be twice as much chance of hitting land as there is hitting water

The map should have a combined area of land that is twice that of the combined area of water.

Students may produce a simple (one land mass, one body of water), complex (many land forms and bodies of water), or somewhere in between

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Exercise 2

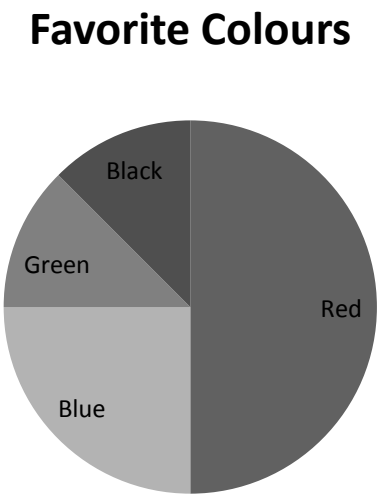
Pie Graphs

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1) Complete the following table

Category	Quantity	Percentage	Degrees of pie chart
1	10	5%	18
2	40	20	72
3	20	10	36
4	50	25	90
5	80	40	144
Total	200	100	360

2) The pie graph below shows the favourite colour of a number of people



a) Sixty people's favourite colour was red. How many people were surveyed?

Red takes up half the pie chart, so the full circle (number of people) must be 120

b) How many people preferred blue?

One quarter of the people, which is 30?

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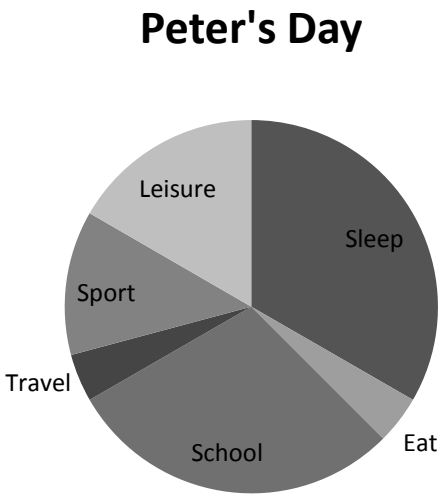
c) Which two colours were preferred by the same number of people?

Green and black (same area of pie chart)

d) Name a colour that was nobody's favourite

Any colour that is not on the pie chart

3) The pie graph shows the amount of time Peter spends doing various things in a 24 hour day



a) Which activities does he spend least time on?

Travelling & eating

b) What fraction of the circle is taken up by sleep?

The sector measures 120°, which is on third of the circle

c) How many hours of sleep does Peter get?

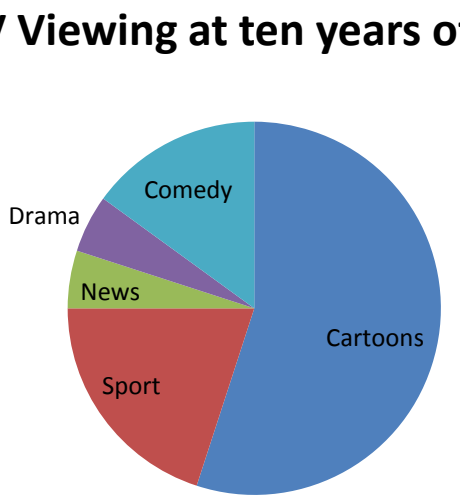
One third of 24 hours equals 8 hours

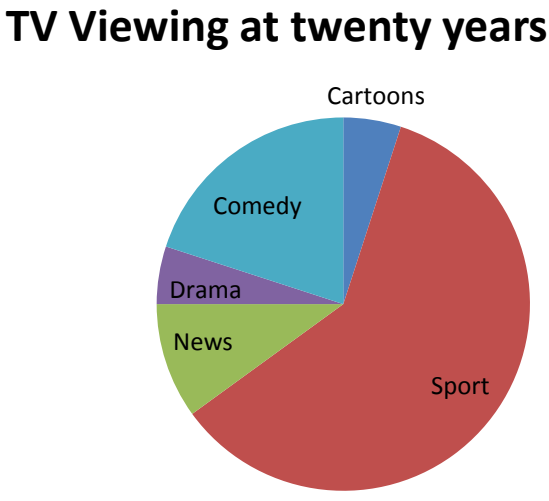
d) Does he spend more time on sport or leisure? Explain your answer

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Sport and leisure take up equal portions of the pie chart, which indicates he spends equal time on them

4) Alan constructed a pie chart when he was ten years old which showed how much time he spent watching certain types of TV programs. He then constructed a similar chart when he was twenty years old





a) Which type of show had the biggest increase in Alan's viewing time?

Sport

b) Which type of show had the biggest decrease in Alan's viewing time?

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Cartoons

c) For which type of show did Alan's viewing habits change the least?

Comedy and drama sectors stayed about the same

d) What percentage of Alan's time was spent watching news when he was ten, and what percentage does he spend now?

At ten years of age the news sector subtended an angle of 30°, which is one twelfth of the pie chart. One twelfth equals 8.5%.

At twenty years of age the news sector subtended an angle of 45°, which is one eighth of the pie chart. One eighth equals 12.5%

e) Explain how the pie graph might change if Alan was to draw one when he was sixty years old

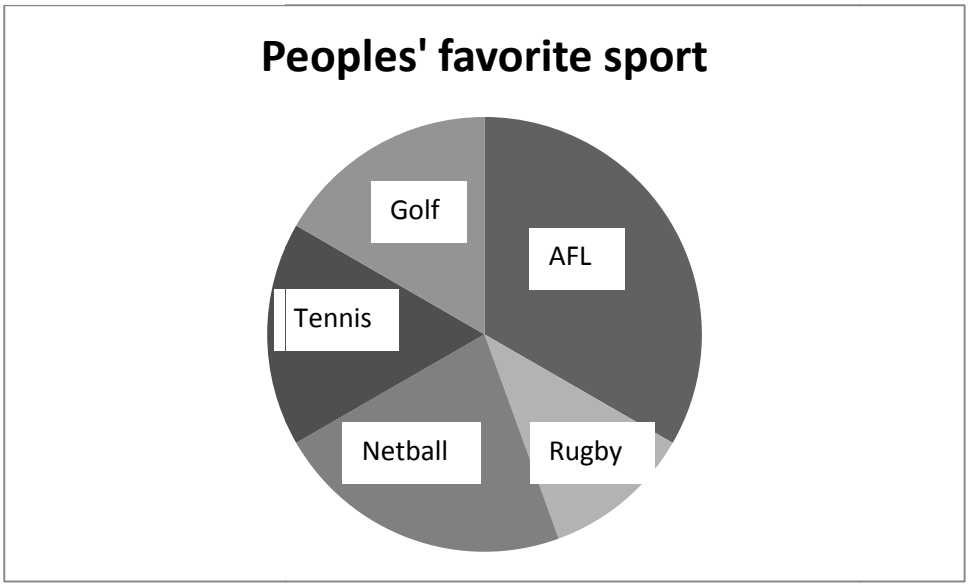
His viewing of cartoons would probably become smaller, whilst the new and drama sectors would probably increase in size.

5) Draw a pie chart that reflects the following information

People's favourite sport

Sport	Number of people
AFL	60
Rugby	20
Netball	40
Tennis	30
Golf	30

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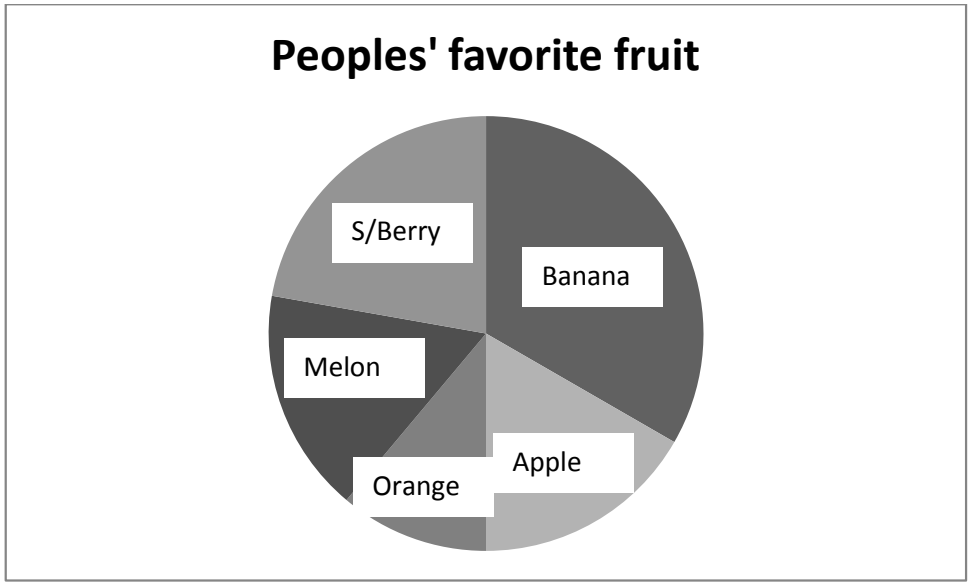


6) A class took a survey of each student's favourite fruit and drew the following graph from their results. One piece of fruit equals one vote



Display this information in a pie chart

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7) Discuss why a pie chart may not be suitable for the following data

WEEK NUMBER	NUMBER OF RAINY DAYS
1	2
2	4
3	0
4	6
5	7
6	4
7	5
8	3
9	2
10	0

There would be too many sectors in the pie graph

Many of the sectors would be the same size: the chart wouldn't show much information

The days when no rain fell would not be shown

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Exercise 3

Mean

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1) Find the mean of the following

a) 2, 2, 2, 2, 2, 2

Sum of scores = 12

Number of scores = 6

Mean $= \frac{12}{6} = 2$

b) 1, 2, 3, 4, 5

Sum of scores = 15

Number of scores = 5

Mean $= \frac{15}{5} = 3$

c) 2, 4, 6, 8, 10

Sum of scores = 30

Number of scores = 5

Mean $= \frac{30}{5} = 6$

d) 10, 20, 30, 40, 50

Sum of scores = 150

Number of scores = 5

Mean $= \frac{150}{5} = 30$

e) 100, 200, 300, 400, 500

Sum of scores = 1500

Number of scores = 5

Mean $= \frac{1500}{5} = 300$

2) Find the mean of the following

a) 5, 6, 7, 8, 9

Sum of scores = 35

Number of scores = 5

Mean $= \frac{35}{5} = 7$

b) 5, 6, 7, 8, 9, 1000

Sum of scores = 1035

Number of scores = 6

Mean $= \frac{1035}{6} = 172.5$

c) 100, 102, 105, 107, 110

Sum of scores = 524

Number of scores = 5

Mean $= \frac{524}{5} = 104.8$

d) 1, 100, 102, 105, 107, 110

Sum of scores = 525

Number of scores = 6

Mean $= \frac{525}{6} = 87.5$

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3) From your answers to question 2, what effect does adding a number greater or less than the rest of the data have on the mean?

The mean value is pushed toward the added number

4) The mean of a set of data is 5. The data is:

1, 3, 7, 8, 9 and an unknown number

What is the unknown number?

There are 6 scores altogether

The mean is 5, so the sum of the data is $6 \times 5 = 30$

The known scores add to 28, so the missing score is 2

5) The mean of the following set of data is 10:

2, 8, 12, 15 and an unknown number

What is the unknown number?

There are 5 scores altogether

The mean is 10, so the sum of the data is $5 \times 10 = 50$

The known scores add to 37, so the missing score is 13

6) Alan played 5 games of basketball and recorded how many points he scored. His scores were:

12, 6, 22, 4, 26

What was his average score per game?

Sum of scores = 70

Number of scores = 5

Mean $= \frac{70}{5} = 14$

7) In his next 5 games Alan averaged 10 points per game. Does this mean that in at least one of his games Alan scored exactly 10 points? Explain your answer

No, as long as the sum of his scores is 50 he will average 10 points per game. There are many combinations of scores that add to 50 that do not include a score of 10

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8) There are 10 boys and 10 girls in the year six maths class. On a test the boys' average score was 70 and the girls' average score was 80. What was the average score for the class?

The sum of the boys' marks was $10 \times 70 = 700$

The sum of the girls' marks was$10 \times 80 = 800$

The total score for the class was 1500

Altogether there were 20 scores

Mean $= \frac{1500}{20} = 75$

9) There are 10 boys and 15 girls in the year seven maths class. On a test the boys' average score was 60 and the girls' average score was 80. What was the average score for the class?

The sum of the boys' marks was $10 \times 60 = 600$

The sum of the girls' marks was $15 \times 80 = 1200$

The total score for the class was 1800

Altogether there were 25 scores

Mean $= \frac{1800}{25} = 72$

10) What happens to the mean of a set of data when every score is multiplied by two? Explain using examples

The mean is also multiplied by 2

Example: The mean of 1, 2, 3, 4, and 5 is 3

The mean of 2, 4, 6, 8, and 10 is 6

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11) What happens to the mean of a set of data when five is added to every score? Explain using examples

The mean also increases by 5

Example: The mean of 1, 2, 3, 4, and 5 is 3

The mean of 6, 7, 8, 9, and 10 is 8

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Year 6 Mathematics

Algebra & Patterns

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Exercise 1

Calculating Values

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1) Calculate the missing values in the following

a) $3 + 7 = 12 - \blacksquare$

2

b) $8 + 9 = 20 - \blacksquare$

3

c) $5 + 11 = 25 - \blacksquare$

9

d) $22 - 8 = 11 + \blacksquare$

3

e) $\blacksquare - 7 = 21 - 9$

19

f) $6 + \blacksquare = 15 - 3$

6

2) Calculate the missing values in the following

a) $42 \div 7 = 3 \times \blacksquare$

2

b) $24 \div 3 = \blacksquare \times 8$

1

c) $6 \times 5 = \blacksquare \div 3$

90

d) $4 \times 10 = 80 \div \blacksquare$

2

e) $1 \times 1 = 8 \div \blacksquare$

8

f) $12 \times \frac{1}{2} = 12 \div \blacksquare$

2

3) Calculate the missing values in the following

a) $4 \times 3 = \blacksquare - 6$

18

b) $7 \times \blacksquare = 14 + 7$

3

c) $22 \div 11 = 12 - \blacksquare$

10

d) $24 \div \blacksquare = 1 \times 4$

6

e) $\blacksquare \times \blacksquare = 4 + 5$

3

f) $300 \times \blacksquare = 45 - 45$

0

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4) Peter thinks of a number and doubles it. After this it is equal to John's number plus 8. If John's original number was 14, what was Peter's original number?

$14 + 8 = 22$ is John's number plus 8

$\square \times 2 = 22$

$\square = 11$

Peter's number was 11

5) Alan's dad says to Alan "If you take your age now and triple it, it will be equal to my age now plus three". If Alan is 12 years old, how old is his dad?

$12 \times 3 = 36$ is Alan's age tripled

$\square + 3 = 36$

$\square = 33$

Alan's dad is 33

6) Greg runs a lap of the track in 4 minutes. In the time it takes him to run three laps, Tom runs two laps. How long does it take Tom to run a lap?

$3 \times 4 = 12$ minutes for Greg to run 3 laps

$\square \times 2 = 12$

$\square = 6$

Tom takes 6 minutes to run a lap

7) If you double the number of behinds a football team kicked and subtract three from the number of goals it kicked, you get the same number. If they kicked 21 goals how many behinds did they kick?

$21 - 3 = 18$

$\square \times 2 = 18$

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$\square = 9$

The team kicked 9 behinds

8) Ben's mother cut up a block of chocolate into equal sized pieces. She cut it so that Ben and his seven friends each got the same amount of chocolate, and there were 2 pieces left over. If each of the children got 4 pieces, how many pieces did she cut the chocolate bar into?

8 children $\times$ 4 pieces $= 32$

$32 + 2 = 34$

There were 34 pieces of chocolate

9) If you multiply the temperature in degrees Celsius by 9, then divide the result by five, the result is the temperature in Fahrenheit minus 32. If the temperature is 40 degrees Celsius, what is the temperature in Fahrenheit?

$40 \times 9 = 360$

$360 \div 5 = 72$

$\square - 32 = 72$

$\square = 104$

40 degrees Celsius equals 104 degrees Fahrenheit

10) One third of light globes in a carton were broken. Of the number left there was enough to put two lights in each of five rooms. How many light globes were originally in the carton?

5 rooms X 2 globes each = 10 globes

One third of the globes were broken, so the good globes must be the other two thirds

$\frac{2}{3}\square = 10$

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$\square = 15$

There were originally 15 globes in the carton

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Exercise 2

Number Patterns

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1) For the following series, fill in the next two terms

a) 1, 3, 7, 15

Multiply by 2 then add 1

31, 63

b) 2, 3, 5, 9

Multiply by 2 then subtract 1

17, 33

c) 1, 4, 13, 40

Multiply by 3 then add 1

121, 364

d) 3, 10, 24

Multiply by 2 then add 4

52, 108

2) For the following series, fill in the next two terms

a) 62, 30, 14

Divide by 2 then subtract 1

6, 4

b) 32, 44, 68, 116

Multiply by 2 then subtract 20

212, 404

c) 100, 60, 40

Divide by 2 then add 10

30, 25

d) 60, 32, 18

Divide by 2 then add 2

11, 7.5

3) Fill in the blanks in the following

a) 2, 7, ___, 67, 202, ___

Multiply by 3 then add1

22, 607

b) ___, 2, 6, 22, 86, ____

Multiply by 4 then subtract 2

1, 342

c) 1, 2, 5, 14, ___

Multiply by 3 then subtract 1

41

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d) 0.5, 1.5, 3.5, ___, ___

Multiply by 2 then add 0.5

7.5, 15.5

e) $\frac{1}{2}, \frac{1}{5}$, ___, $\frac{1}{23}$, ___

Multiply the denominator by 2 then add 1

$\frac{1}{11}, \frac{1}{47}$

4) What are the next three numbers of the following series?

1, 3, 6, 10

The difference between the numbers is one more than the previous difference

15, 21

5) Each day Peter decided to walk double the number of steps he had taken the day before plus 10 extra steps. On the first day he walked 10 steps

a) How many steps did he take on day 2?

Day 2 walked $2 \times 10 + 10 = 30$ steps

b) How many steps did he walk on day 5?

Day 3 walked $30 \times 2 + 10 = 70$ steps

Day 4 walked $70 \times 2 + 10 = 150$ steps

Day 5 walked $150 \times 2 + 10 = 310$ steps

c) How many steps had he taken at the end of the week?

Day 6 walked $310 \times 2 + 10 = 630$ steps

Day 7 walked $630 \times 2 + 10 = 1270$ steps

Total steps for the week was $10 + 30 + 70 + 150 + 310 + 630 + 1270 = 2470$ steps

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6) On week one of his diet, George lost .2 kg. Each week after that he lost double the previous weeks minus 1kg.

a) How much weight did he lose in week 3?

In week two he lost $2 \times 2 - 1 = 3$kg

In week three he lost $3 \times 2 - 1 = 5$kg

b) After five weeks he had reached his goal weight. What was his average weight loss per week?

In week four he lost $5 \times 2 - 1 = 9$kg

In week five he lost $9 \times 2 - 1 = 17$kg

In the five weeks he lost $2 + 3 + 5 + 9 + 17 = 36$kg

The average weight lost was $\frac{36}{5} = 7.2$kg

7) A pond can only hold 100 fish until it has to be cleaned out. In the first month there were 4 fish in the pond. Each month the number of fish in the pond was equal to three times the previous months' number less 6. During which month was it necessary to clean out the fish pond, and how many fish would have been in the pond at the end of that month if it had not been cleaned out?

Month 1 there were 4 fish

Month 2 there were $4 \times 3 - 6 = 6$ fish

Month 3 there were $6 \times 3 - 6 = 12$ fish

Month 4 there were $12 \times 3 - 6 = 30$ fish

Month 5 there were $30 \times 3 - 6 = 84$ fish

Month 6 there would have been $84 \times 3 - 6 = 246$ fish

Therefore the pond needed to be cleared in the sixth month

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Year 6 Mathematics

Measurement:

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Exercise 1

Distance & Travel

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1) Calculate the distance around the outside of a square that has a side length of

a) 3 mm

$4 \times 3 = 12\text{mm}$

b) 8 cm

$4 \times 8 = 32\text{cm}$

c) 32 m

$4 \times 32 = 128\text{m}$

d) 5.4 km

$4 \times 5.4 = 21.6\text{km}$

e) 70 cm

$4 \times 70 = 280\text{cm}$

2) Calculate the distance around the outside of a rectangle that has side lengths of

a) 4m and 3 m

$2 \times (4 + 3) = 14\text{m}$

b) 17mm and 10mm

$2 \times (17 + 10) = 54\text{mm}$

c) 4 m and 50cm

50cm =0.5m

$2 \times (4 + 0.5) = 9\text{m}$

d) 0.8 km and 60cm

0.8 km = 800 m

60 cm = 0.6 m

$2 \times (800 + 0.6) = 1601.2\text{m}$

e) 90 mm and 9cm

90 mm = 9 cm

$2 \times (9 + 9) = 36$ cm

3) Calculate the distance around the outside of an equilateral triangle of side length

a) 9 cm

$3 \times 9 = 27$ cm

b) 3 m

$3 \times 3 = 9$ m

c) 1.2cm

$3 \times 1.2 = 3.6$ cm

d) 4 km

$3 \times 4 = 12$ km

e) 1.2 m

$3 \times 1.2 = 3.6$ m

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4) Calculate the distance around the outside of an isosceles triangle with the following measurements

a) 10cm for the common sides, and 5cm for the unique side

$(2 \times 10) + 5 = 15$ cm

b) 100 cm for the common sides, and 5cm for the unique side

$(2 \times 100) + 5 = 205$ cm

c) 10cm for the common sides, and 15cm for the unique side

$(2 \times 10) + 15 = 35$ cm

d) 30cm for the common sides, and 8cm for the unique side

$(2 \times 30) + 8 = 68$ cm

e) 25mm for the common sides, and 1cm for the unique side

$(2 \times 25) + 1 = 51$ cm

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5) A car travels at an average speed of 100 km per hour. How long would it take to complete a journey of 300 km?

$300km = 3 \times 100km$

The car will take 3 hours to go 300 km

6) A man walks at an average speed of 5 km per hour. How long would it take him to walk 7.5 km?

$7.5km = 1.5 \times 5km$

The man will take 1.5 hours to go 7.5 km

7) How long would it take an ant that walks at an average speed of 2cm per minute to walk around a square of side length 5cm?

The perimeter of the square is $4 \times 5cm = 20cm$

$20cm = 10 \times 2cm$

The ant will take 10 minutes to walk around the square

8) An equilateral triangle has a side length of 8 cm. It took a snail one hour to walk around the triangle. What was its average speed?

The perimeter of the triangle is $3 \times 8cm = 24cm$

Average speed was 24 cm in 1 hour = 24 cm per hour

9) A rectangle has one side that measures 45 mm and another that measures 2cm. How long would it take to go around its perimeter if the average speed was 1.3 cm per minute?

$45mm = 4.5cm$

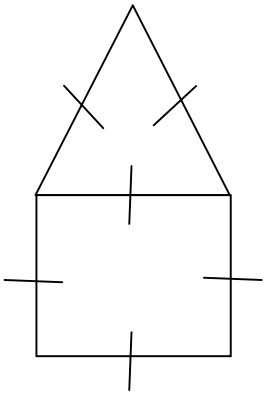
The perimeter of the rectangle is $2 + 4.5 + 2 + 4.5 = 13cm$

$13cm = 10 \times 1.3cm$

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The snail took 10 minutes to crawl around the triangle

10) How long would it take a man walking at 2 km per hour to go around the outside of this shape?



The perimeter of the shape is $4 + 4 + 4 + 4 + 4 = 20km$

$20km = 10 \times 2km$

It takes the man 10 hours to walk around the shape

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Exercise 2

Area

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Use the following scale to calculate the area of the following shapes

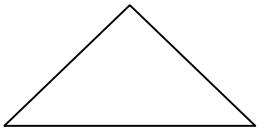
---3 cm---

1)



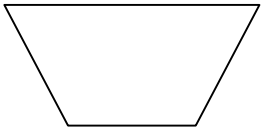
The rectangle is 3cm wide by 6cm long. The area is $6 \times 3 = 18cm^2$

2)



Base of triangle is 6 cm, height is 3 cm. The area is $\frac{1}{2} \times 6 \times 3 = 9cm^2$

3)



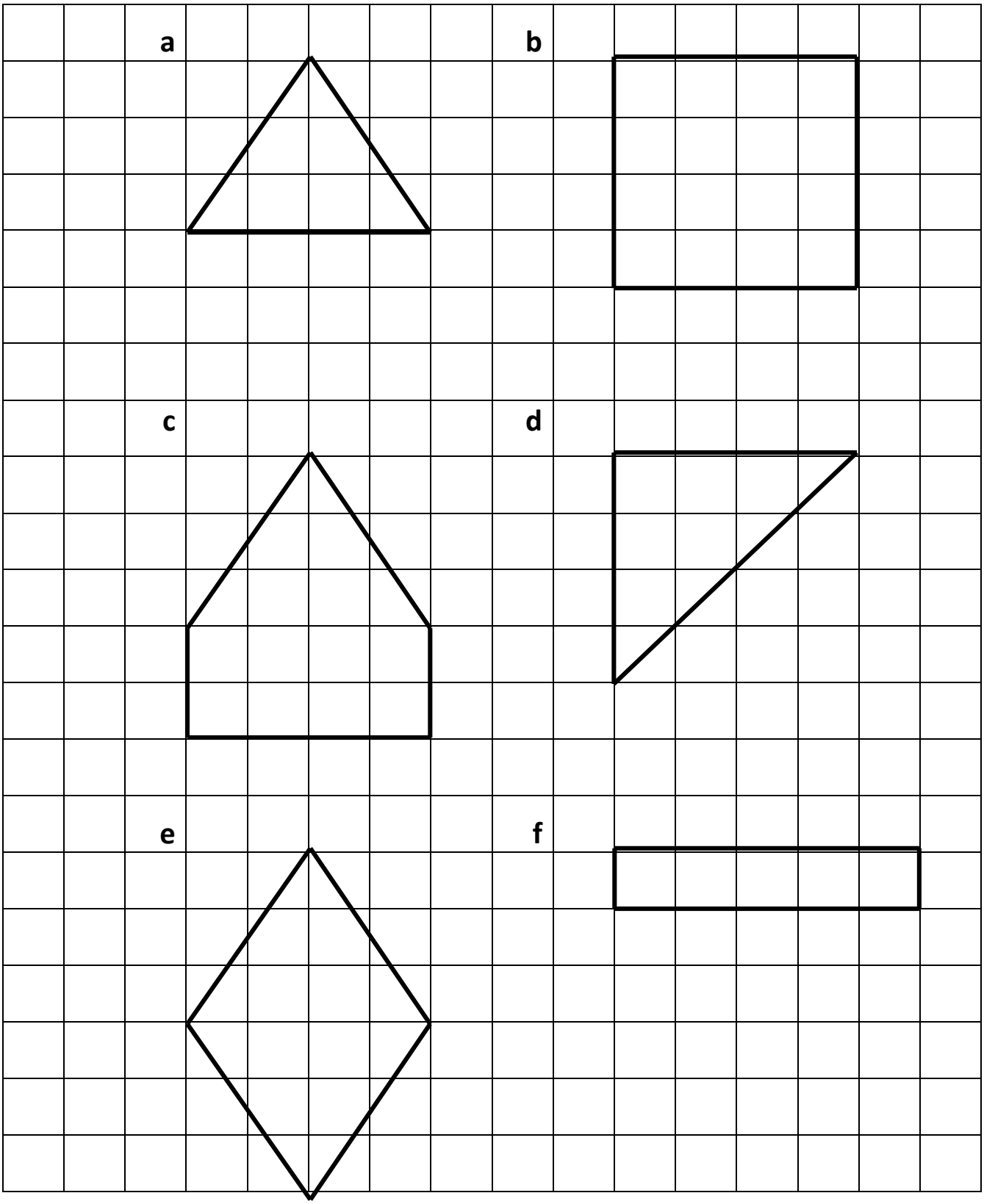
The trapezium is 3 cm high; its two sides are 6 cm and 3 cm long

The area is $3 \times \frac{6+3}{2} = 13.5cm^2$

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4)

Use the scale 1 cm = 4 metres to calculate the areas of the following



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a) $\frac{1}{2} \times 16 \times 12 = 96m^2$

b) $16 \times 16 = 256m^2$

c) $(16 \times 8) + \left(\frac{1}{2} \times 16 \times 12\right) = 224m^2$

d) $\frac{1}{2} \times 16 \times 16 = 128m^2$

e) $\frac{1}{2} \times 24 \times 16 = 192m^2$

f) $20 \times 4 = 80m^2$

5) A man takes 200 paces around a square. If each pace is one meter, what is the area of the square?

Perimeter of square is 200 m; therefore the length of each side is 50m

Area $= 50 \times 50 = 2500m^2$

6) A drawing is scaled down in the ratio 1:10. On the drawing the sides of a rectangle are 10 cm by 15 cm. What is the area of the full size rectangle?

The actual size of the rectangle is (10×10) by (10×15) =100 cm by 150 cm

Area $= 100cm \times 150cm = 1m \times 1.5m = 1.5m^2$

7) The area of a square is 2500 cm^2. If the sides of the square are reduced to one fifth their size, what is the area of the new square?

A square of area 2500 cm^2 has a side length of 50 cm. $50 \times \frac{1}{5} = 10cm$

Area of new square $10 \times 10 = 100cm^2$

8) A movie projector blows the picture on a screen up to one hundred times its original size. What area would a triangle of base length 2 cm and height 5 cm have on the projector screen?

Area of actual triangle $= \frac{1}{2} \times 2 \times 5 = 5cm^2$

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Area of triangle on screen $= 5 \times 100 = 500cm^2$

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Exercise 3

Volume & Capacity

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1) How much water is displaced from a container if a cube with the following side lengths is placed into it?

a) 10 cm

Volume of cube $= 1000cm^3$

$1000cm^3 = 1$ litre

b) 20 cm

Volume of cube $= 8000cm^3$

$8000cm^3 = 8$ litres

c) 40 cm

Volume of cube $= 64000cm^3$

$64000cm^3 = 64$ litres

d) 1 m

Volume of cube $1m^3$

$1m^3 = 1000$ litres

2) What is the side length of a cube that displaces the following amount of water?

a) 125 litres

125 litres is displaced by a volume of $125000cm^3$

A cube with a volume of $125000cm^3$ has a side length of $50cm$

b) 8 litres

8 litres is displaced by a volume of $8000cm^3$

A cube with a volume of $8000cm^3$ has a side length of $20cm$

c) 27 litres

27 litres is displaced by a volume of $27000cm^3$

A cube with a volume of $27000cm^3$ has a side length of $30cm$

d) 1000 litres

1000 litres is displaced by a volume of $1000000cm^3$

A cube with a volume of $1000000cm^3$ has a side length of $100cm = 1m$

3) How much water is displaced from a container if a cube with the following side lengths is placed into it?

a) 1 cm

Volume of cube $= 1cm^3$

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$1cm^3 = 1$ mL

b) 7 cm

Volume of cube $= 343cm^3$

$343cm^3 = 343$ mL

c) 5 cm

Volume of cube $= 125cm^3$

$125cm^3 = 125$ mL

d) 3 cm

Volume of cube $= 27cm^3$

$27cm^3 = 27$ mL

4) What is the side length of a cube that displaces the following amount of water?

a) 1 mL

1 mL is displaced by a volume of $1cm^3$

A cube with a volume of $1cm^3$ has a side length of $1cm$

b) 64 mL

64 mL is displaced by a volume of $64cm^3$

A cube with a volume of $64cm^3$ has a side length of $4cm$

c) 512 mL

512 mL is displaced by a volume of $512cm^3$

A cube with a volume of $512cm^3$ has a side length of $8cm$

d) 27 mL

27 mL is displaced by a volume of $27cm^3$

A cube with a volume of $27cm^3$ has a side length of $3cm$

5) A rock having a volume of 805 cubic centimetres is placed in a full tub of bath water. How much water is displaced?

$805cm^3$ displaces 805 mL of water

6) Seven litres of water is displaced from a full swimming pool when a large object is placed into it. What was the volume of the object?

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7 litres is displaced by a volume of $7000cm^3$

7) Archimedes reputedly was the first person to realise the relationship between the volume of a body being placed in water and the amount of water that is displaced. He is alleged to have been so excited that he immediately leapt from his bath shouting "Eureka" when he realised his discovery. If Archimedes had a volume of 0.095 m^3, how much water would he have displaced from his bath?

$0.095m^3$ displaces 95 litres

8) Convert the following to litres

a) 1400 mL

$\frac{1400}{1000} = 1.4$ litres

b) 3500 mL

$\frac{3500}{1000} = 3.5$ litres

c) 2225 mL

$\frac{2225}{1000} = 2.225$ litres

d) 3950 mL

$\frac{3950}{1000} = 3.95$ litres

e) 5010 mL

$\frac{5010}{1000} = 5.01$ litres

9) Convert the following to mLs

a) 2.12 litres

$2.12 \times 1000 = 2120$ mL

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b) 1.356 litres

$1.356 \times 1000 = 1356$ mL

c) 0.1 litres

$0.1 \times 1000 = 100$ mL

d) 0.87 litres

$0.87 \times 1000 = 870$ mL

e) 10.01 litres

$10.01 \times 1000 = 10010$ mL

f) 0.01 litres

$0.01 \times 1000 = 10$ mL

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Exercise 4

Mass

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1) A sack of identical rocks weighs 54 kg. If there are 9 rocks in the bag, what is the mass of each rock?

$54 \div 9 = 6$ kg each

2) A packet of lollies weighs 400 g. If there are 20 lollies in the pack, how much does each lolly weigh?

$400 \div 20 = 20$ grams each

3) The mass of an empty container is 40 grams. Into the container there are placed 15 identical wooden blocks. The container is put on a scale and it reads 265 grams. How much does each block weigh?

$265 - 40 = 225$ grams is mass of blocks only

$225 \div 15 = 15$ grams each

4) A container has a mass of 20 grams. When some identical pens are put into it the mass increases to 220 grams. If each pen weighs 40 grams, how many pens are placed in the container?

$220 - 20 = 200$ grams is the mass of the pens

$200 \div 40 = 5$ pens

5) A container holding twelve identical pieces of plastic each weighing 11 grams is placed on a scale which reads 170 grams what is the mass of the container?

$12 \times 11 = 132$ grams is the mass of the plastic pieces

$170 - 132 = 38$ grams is the mass of the container without the pens

6) Convert the following to grams

a) 1 kg

$1 \times 1000 = 1000$ g

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b) 4 kg

$4 \times 1000 = 4000$ g

c) 6.5 kg

$6.5 \times 1000 = 6500$ g

d) 3.25 kg

$3.25 \times 1000 = 3250$ g

e) 0.932 kg

$0.932 \times 1000 = 932$ g

f) 1.604 kg

$1.604 \times 1000 = 1604$ g

7) Convert the following to kg

a) 1000 g

$1000 \div 1000 = 1$ kg

b) 3000 g

$3000 \div 1000 = 3$ kg

c) 2500 g

$2500 \div 1000 = 2.5$ kg

d) 875 g

$875 \div 1000 = 0.875$ kg

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e) 10001 g

$10001 = 1000 = 10.001$ kg

f) 203 g

$203 \div 1000 = 0.203$ kg

8) What is the mass of the following quantities of water?

(1 litre of water has a mass of 1 kg)

a) 1 litre

1 kg

b) 5 litres

5 kg

c) 3.5 litres

3.5 kg

d) 0.75 litres

0.75 kg = 750 g

e) 400 mL

0.4 kg = 400 mL

f) 2500 mL

2.5 kg

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9) What is the volume of the following masses of water?

a) 1 kg

1 litre

b) 4 kg

4 litres

c) 2.5 kg

2.5 litres

d) 0.6 kg

600 mL

e) 300 g

300 mL

f) 4250 g

4.25 litres

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Exercise 5

Time

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1) The following shows Peter's school timetable

	Monday	Tuesday	Wednesday	Thursday	Friday
9 – 9:55	Maths	Science	Option 1	SOCE	Option 2
10-10:55	English	Science	Maths	Science	Option 2
11-11:55	Computing	Option 2	Maths	Computing	English
12-12:35	Lunch	Lunch	Lunch	Lunch	Lunch
12:40-1:35	SOCE	Maths	Science	Maths	SOCE
1:40-2:35	SOCE	SOCE	English	Health	Sport
2:40-3:35	Option 1	English	English	Health	Sport

a) How long does each class last for?

55 minutes

b) How many hours of computing does peter have per week?

110 minutes = 1 hour 50 minutes

c) Peter chose woodwork as his second option. When does he have woodwork class?

Friday at 9 am

d) Peter likes Maths & English but doesn't like SOCE. Which day is probably his favourite day at school?

Wednesday

e) How many subjects have "double periods"?

SOCE, Science, Maths, English, Health, Option 2, Sport

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2) The following timetable shows the places and departure and arrival times of free buses to "The Big Day Out" music festival

FREE BUS TIMETABLE

BUS 1

Ventnor, Boots Bus Stop	10:00 Depart	Freshwater, Big Day Out	15:30 Depart
Niton, Church Street	10:20	Brighstone, School	15:50
Brighstone, School	10:40	Niton, Church Street	16:10
Freshwater, Big Day Out	11:00 Arrive	Ventnor, Boots Bus Stop	16:30 Arrive

BUS 2

Bembridge Library	10:00 Depart	Freshwater, Big Day Out	15:30 Depart
Sandown, High Street	10:20	Sandown, High Street	16:20
Freshwater, Big Day Out	11:10 Arrive	Bembridge Library	16.40 Arrive

BUS 3

Shanklin, Bus Station	10:00 Depart	Freshwater, Big Day Out	15:30 Depart
Freshwater, Big Day Out	11:00 Arrive	Shanklin, Bus Station	16.30 Arrive

BUS 4

Ryde Commodore	10:00 10:30 Depart	Freshwater, Big Day Out	15:30 16:00 Depart
Freshwater, Big Day Out	11:00 11:30 Arrive	Ryde Commodore	16.30 17:00 Arrive

BUS 5

Cowes Library	10:00 Depart	Freshwater, Big Day Out	15:30 Depart
Freshwater, Big Day Out	10:50 Arrive	Cowes Library	16.20 Arrive

BUS 6

Newport Library	10:00 10:30 Depart	Freshwater, Big Day Out	15:30 16:00 Depart
Freshwater, Big Day Out	10:30 11:00 Arrive	Newport Library	16.00 16:30 Arrive

BUS 7

East Cowes Town Hall	10:00 Depart	Freshwater, Big Day Out	15:30 Depart
Freshwater, Big Day Out	10:45 Arrive	East Cowes Town Hall	16.15 Arrive

a) How many buses are there?

6

b) Which bus runs from Cowes Library?

Bus 5

c) How many places does bus 1 pickup from?

Three

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d) How long does the trip from The Big day Out to East Cowes Town Hall take?

45 minutes (bus 7)

e) How long does it take to get from Sandown to Bembridge Library?

20 minutes (bus 2)

3) The timetable for three trains from Midland to Perth is shown below

	a.m.	p.m.	
Midland	8:35	2:28	4:06
West Midland	Exp	2:30	4:08
East Guildford	Exp	2:32	4:10
Guildford	Exp	2:33	4:11
Success Hill	Exp	2:34	4:12
Bassendean	8:40	2:36	4:14
Ashfield	Exp	2:38	4:16
Bayswater	8:43	2:41	4:19
Meltham	8:45	2:43	Exp
Maylands	8:46	2:44	Exp
Mt. Lawley	8:48	2:46	Exp
East Perth	8:50	2:48	Exp
Claisebrook	8:51	2:49	4:23
McIver	8:53	2:51	4:25
Perth	8:55	3:00	4:27

Exp = Express

a) Which train gets you to Perth the quickest?

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The train that leaves at 8:35 takes 20 minutes

b) How long does it take to get from Bassendean to East Perth on a non express train?

12 minutes (second train)

c) How much time is saved between Bayswater and McIver if the express train is taken?

Non express train takes 10 minutes; express train takes 6 minutes, saving 4 minutes

d) Name a station that all three trains stop at

Bassendean, Bayswater, Claisebrook, McIver, Perth

e) How long does an express train take to get from Midland to Bassendean?

5 minutes

4) The timetable below shows air arrivals and departures to and from Gibraltar

Information kindly supplied by Brian T Richards.

Day	Flight No.	Airline	Arr.	From	Dep.	Flight No.	To
Monday	ZB068	Monarch	11.00	Luton	12.00	ZB069	Luton
	ZB574	Monarch	11.15	Manchester	12.00	ZB575	Luton
	BA490	British Airways	11.55	Heathrow	12.45	BA491	Heathrow
	EZY8903	easyJet	13.50	Gatwick	14.25	EZY8904	Gatwick
Tuesday	ZB068	Monarch	11.55	Luton	12.40	ZB069	Luton
	BA490	British Airways	11.55	Heathrow	12.45	BA491	Heathrow
	EZY8903	easyJet	13.40	Gatwick	14.10	EZY8904	Gatwick
Wednesday	ZB574	Monarch	11.00	Manchester	11.45	ZB575	Manchester
	ZB068	Monarch	11.00	Luton	12.00	ZB069	Luton
	BA490	British Airways	11.55	Heathrow	12.45	BA491	Heathrow
	EZY8903	easyJet	13.50	Gatwick	14.25	EZY8904	Gatwick
Thursday	ZB068	Monarch	10.40	Luton	11.30	ZB069	Luton
	BA490	British Airways	11.55	Heathrow	12.45	BA491	Heathrow
	EZY8903	easyJet	13.50	Gatwick	14.25	EZY8904	Gatwick
Friday	ZB574	Monarch	11.00	Manchester	11.45	ZB575	Manchester
	BA490	British Airways	11.55	Heathrow	12.45	BA491	Heathrow
	EZY8903	easyJet	13.45	Gatwick	14.25	EZY8904	Gatwick
	ZB062	Monarch	18.55	Luton	19.55	ZB063	Luton
Saturday	EZY8901	easyJet	11.05	Gatwick	11.40	EZY8902	Gatwick
	ZB068	Monarch	11.55	Luton	12.40	ZB069	Luton
	BA490	British Airways	11.55	Heathrow	12.45	BA491	Heathrow
Sunday	EZY8901	easyJet	11.05	Gatwick	11.40	EZY8902	Gatwick
	BA490	British Airways	11.55	Heathrow	12.45	BA491	Heathrow
	ZB062	Monarch	18.10	Luton	18.55	ZB063	Luton

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a) What time does the plane to Gatwick leave on Tuesday?

14:10

b) How many times a week does easy Jet fly into Gibraltar?

7

c) Name an airport that Monarch does not fly into

Heathrow, Gatwick

d) Which airline uses Heathrow?

British Airways

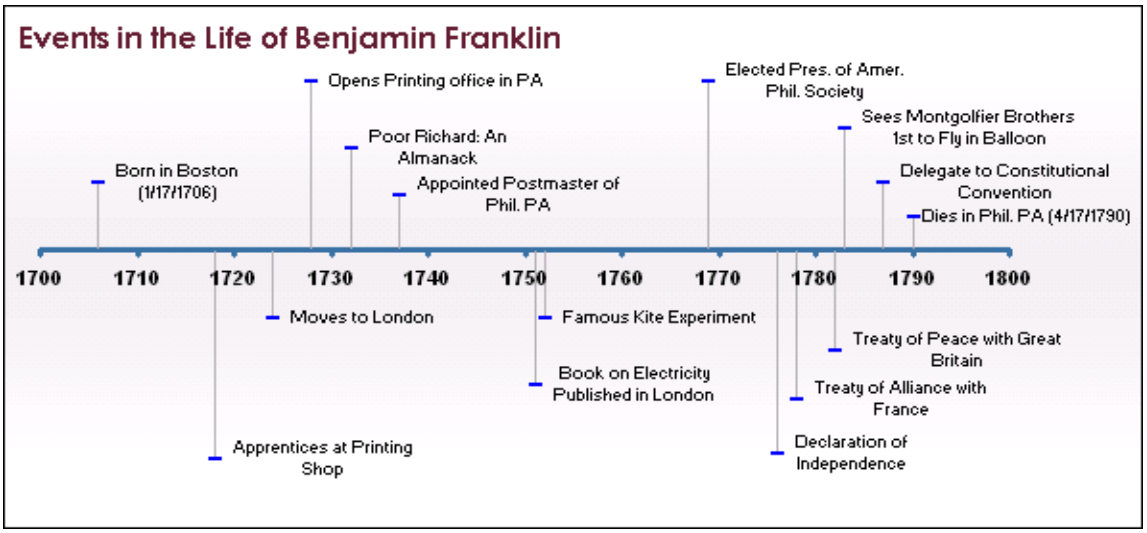
e) On Friday how long does the flight that lands at 13:45 sit on the tarmac before it takes off again?

40 minutes (14:25)

f) How many times a week does flight ZB575 fly out of Gibraltar?

3

5) The timeline shows the major events in the life on Benjamin Franklin



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a) How long did Benjamin Franklin live for?

1790 – 1706 = 84

b) In approximately what year did he move to London?

1725

c) How old was he when the Declaration of Independence was signed?

Approximately 70

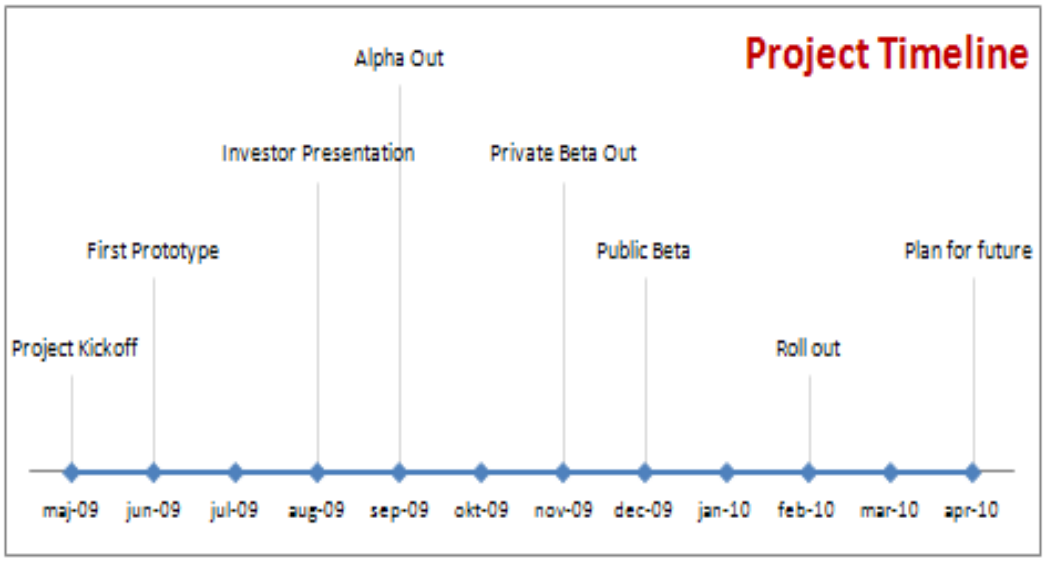
d) Approximately how many years after he published "Poor Richard" did he perform his famous kite experiment?

Approximately 20 years

e) In approximately what year was he elected President of the American Philosophical Society?

1768

6) A project is planned from start to finish by the timeline below



a) When did the project start?

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May 09

b) How long from start up to roll out?

9 months (May 09 to Feb 10)

c) When was the investor presentation?

August 09

d) How long did the project allow for development of the first prototype?

1 month

e) How long after the roll out were future plans made?

2 months (April 10)

7) Construct a timeline for a typical day from 6am to 8pm for a year 6 student on a school day

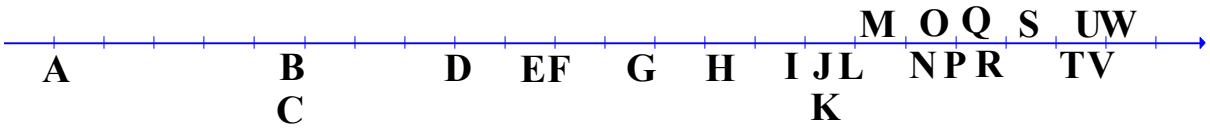
8) Put the following events on a suitably scaled timeline

Technology Timeline 1400-1890

A1450-Printing Press Invented
B1543-Copernicus Claims Earth Circles the Sun
C1543-First Illustrated Study of Human Body
D1609-Kepler Publishes His Laws of Planetary Motion
E1638-Galileo Explains Principles of Falling Bodies
F1669-Newton Explains Calculus
G1698 Steam Pump Introduced
H1733 Flying Shuttle Invented
I1764-Spinng Jenny Introduced
J1776-First Submarine Launched
K1776-Watt Builds Steam Engine
L1784 Power Loom Invented
M1793-Cotton Gin Invented

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N1814-George Stephenson Develop Steam Engine
O1819-First Steam Powered Ship Crosses Atlantic
P1829-Baltimore and Ohio Railroad Begun
Q1837-Steel Plow Invented
R1839-First Photo
S1860-Work Begun on London Underground Railroad
T1876-Telephone Invented
U1879-Electric Light Invented
V1888-George Eastman Patents Camera
W1890-Eiffel Tower



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Year 6 Mathematics

Space

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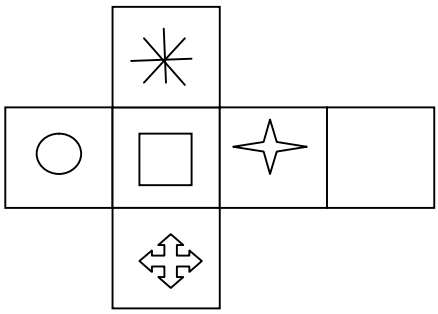
Exercise 1

Representation of 3D Shapes

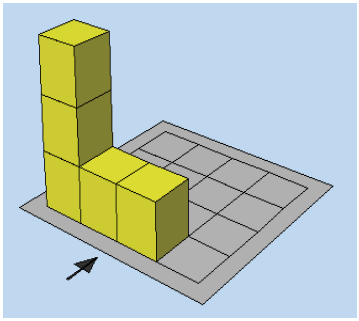
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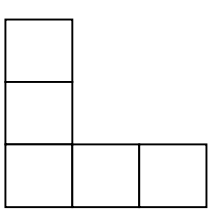
1) Draw the net of the cube shown below

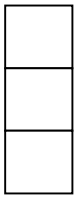




2) Draw the front and two side views of the following shape (the arrow indicates the front)



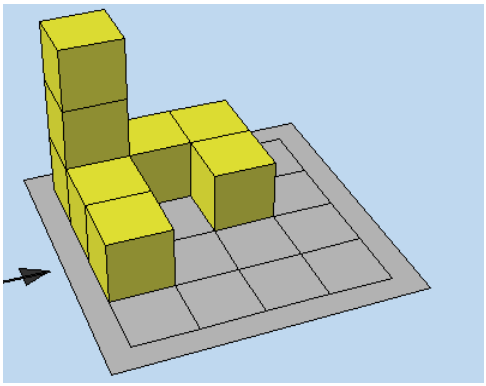


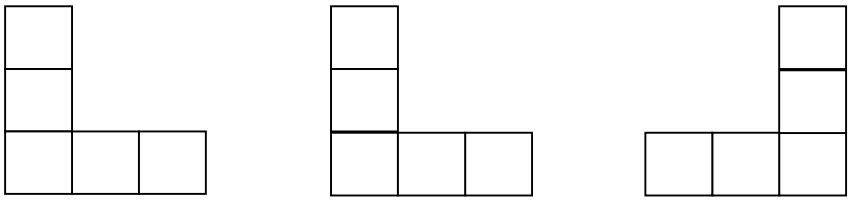




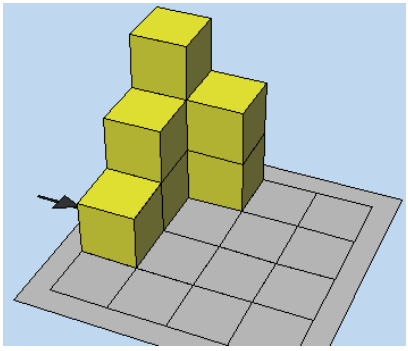
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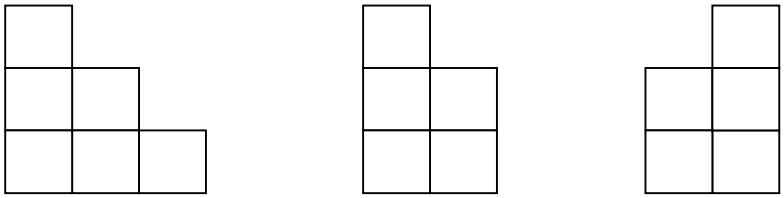
3) Draw the front and two side views of the following shape (the arrow indicates the front)





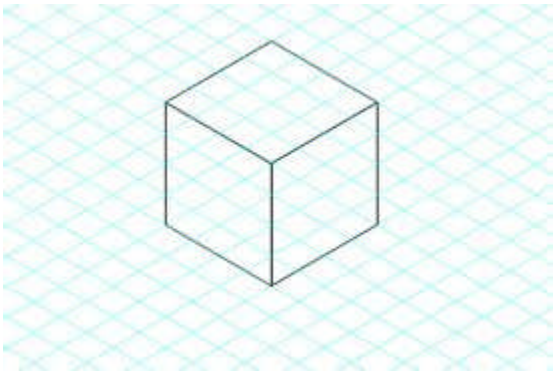
4) Draw the front and two side views of the following shape (the arrow indicates the front



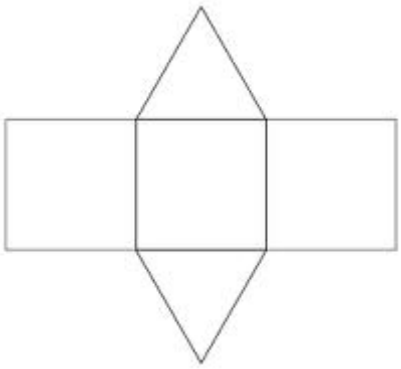


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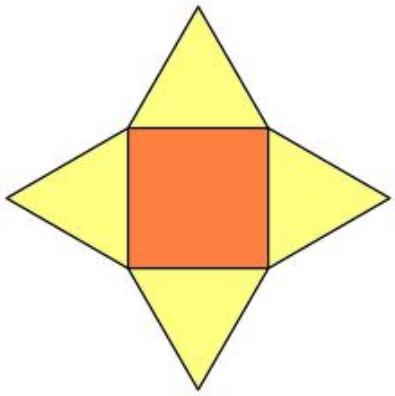
5) Show the isometric representation of a cube



6) Draw the net of a triangular prism



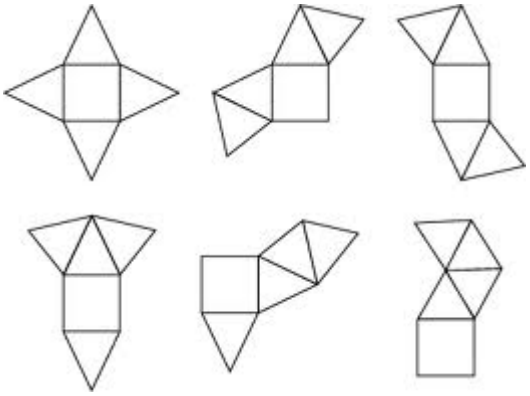
7) The following is the net of which shape?



Square based pyramid

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8) Draw the six possible nets for a square pyramid



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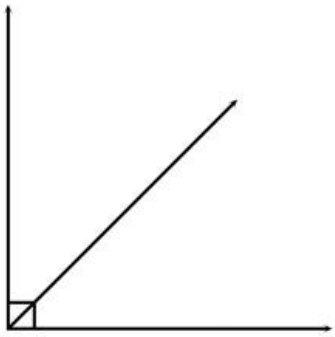
Exercise 2

Angles

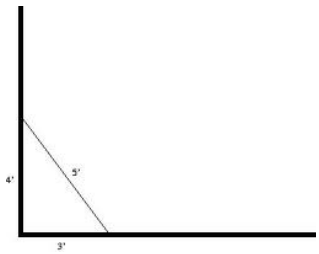
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1) Draw the following angles

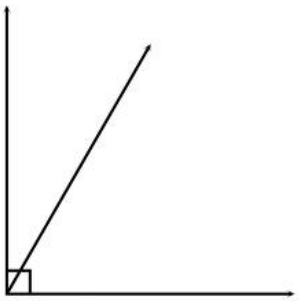
a) 45°



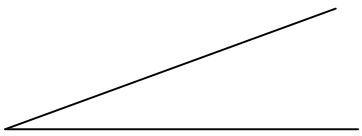
b) 90°



c) 60°

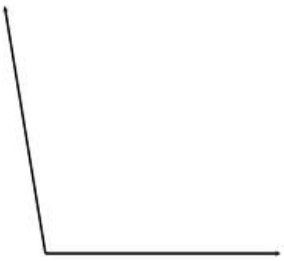


d) 30°

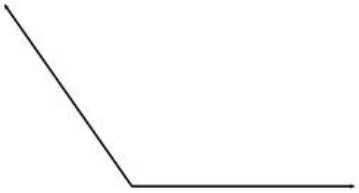


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e) 100°



f) 125°

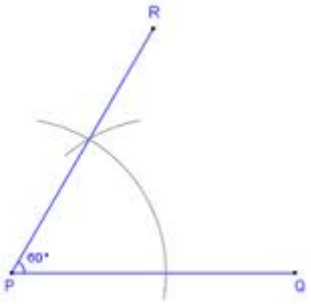


g) 180°



2) Measure the following angles

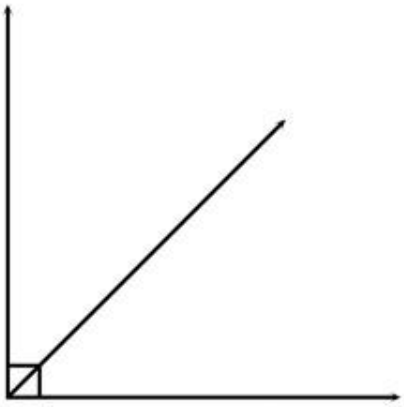
a)



60 degrees

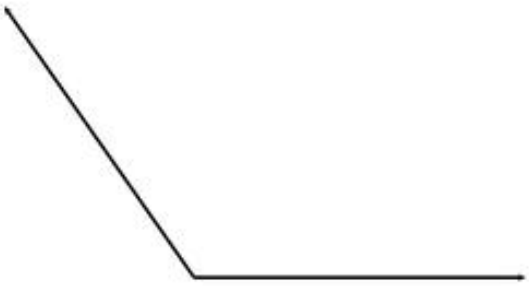
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b)



45 degrees

c)



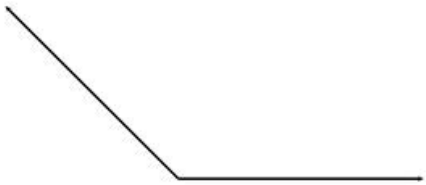
125 degrees

d)

180 degrees

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e)



135 degrees

3) Classify the following angles

a) 30°

Acute

b) 123°

Obtuse

c) 180°

Straight

d) 10°

Acute

e) 90°

Right

f) 359°

Reflex

g) 147°

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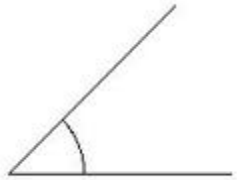
Obtuse

h) 360°

Revolution

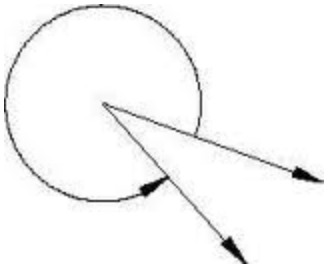
4) Identify the angle type formed by the intersection of the lines

a)



Acute

b)



Acute

c)



Straight

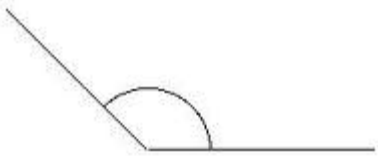
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d)



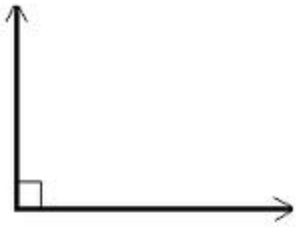
Acute

e)



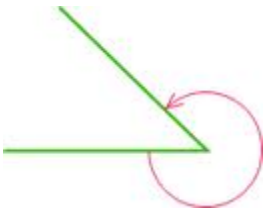
Obtuse

f)



Right

g)



Acute

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5) What is the largest (whole number) value that can be added to each of the following angles so that they remain acute?

a) 30°

59

b) 42°

47

c) 10°

79

d) 57°

32

e) 88°

1

f) 0°

89

6) What is the smallest (whole number) value that can be added to the following angles to make them obtuse?

a) 26°

65

b) 11°

80

c) 0°

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91

d) 89°

2

e) 92°

Already obtuse

7) What is the smallest (whole number) value that can be added to the following angles to make them reflex?

a) 23°

158

b) 93°

88

c) 0°

181

d) 126°

55

e) 178°

3

f) 222°

Already reflex

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8) What is the largest (whole number) value that can be added to the following angles to make the largest obtuse angle possible?

a) 23°

156

b) 93°

86

c) 0°

179

d) 126°

53

e) 178°

1

f) 22°

157

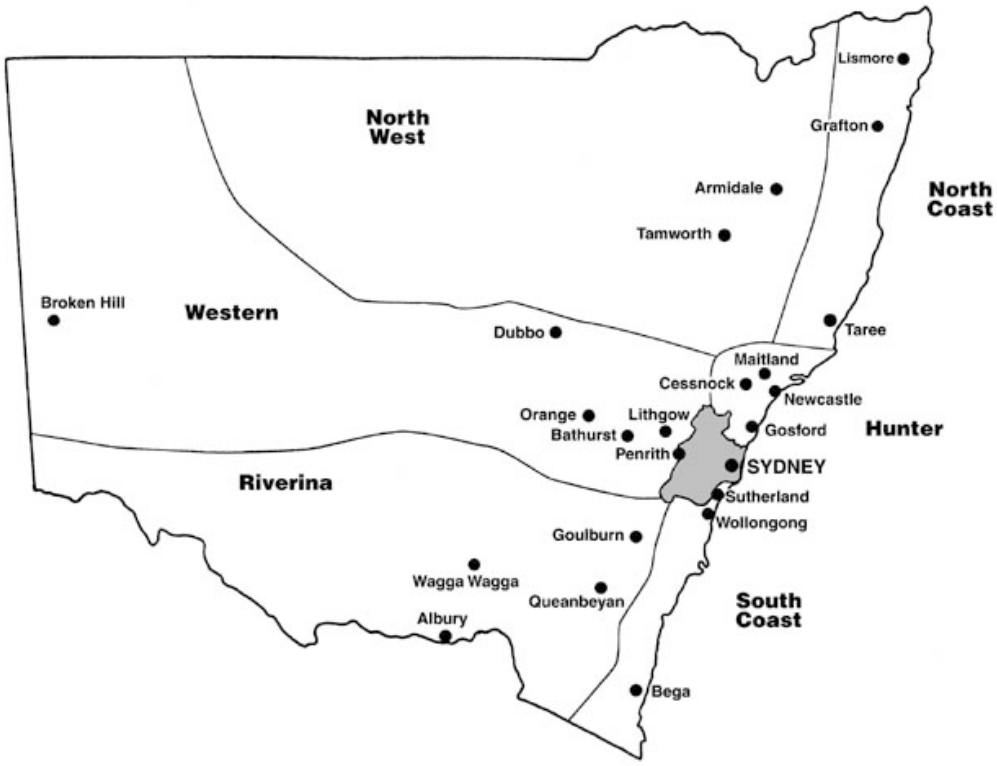
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Exercise 3

Using Maps

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1) Use the map of NSW below to answer the following questions



a) Which direction is Lismore from Grafton?

Close to north

b) Which is the westernmost town on the map?

Broken Hill

c) Name two towns in the Riverina district

Albury, Wagga Wagga, Queanbeyan, Goulbourn

d) Which town is closest to Orange?

Bathurst

e) Victoria is south of NSW; which town lies on the border of the two states?

Albury

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2) Answer the questions from the following map of Trinidad & Tobago



a) Which of the two islands is bigger?

Trinidad

b) Approximately how wide is Trinidad?

40 km at its widest point, approximately 30 km through the centre

c) What is the most north easterly town of Trinidad?

Toco

d) What sea separates the two islands?

Caribbean Sea

e) What is the approximate distance from Port of Spain to Scarborough?

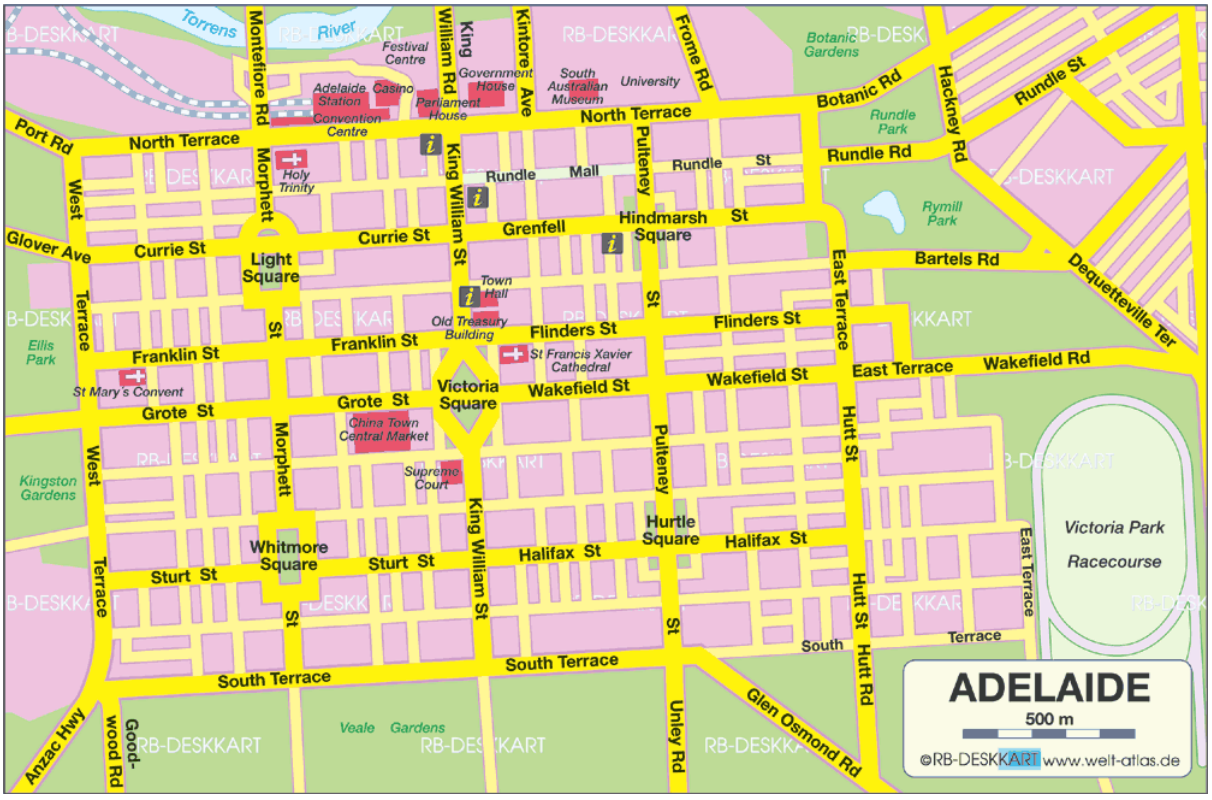
Approximately 60 km

f) What town lies due south of Sangre Grande?

Rio Chiaro

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3) Use the street map of Adelaide city to answer the questions



a) What direction is Light Square from Victoria Square?

North West

b) How long is South Terrace?

Approximately 1750 m (1.75 km)

c) Franklin Street becomes what street?

Flinders Street

d) What is the approximate walking distance from the corner of Halifax and King William streets to Whitmore Square?

Approximately 500 m

e) Which direction does Glen Osmond drive run?

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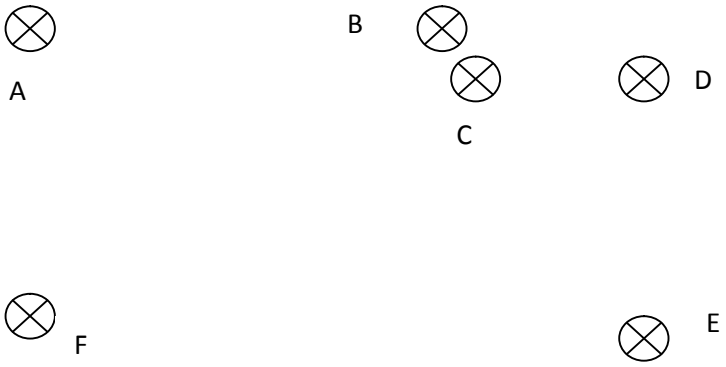
North West to South east

f) At which end of Adelaide is Victoria Park racecourse?

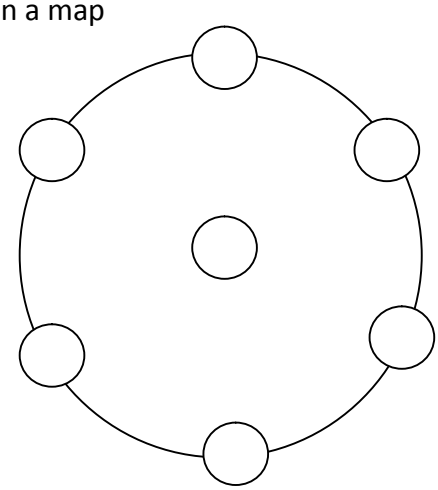
The eastern end

4) Using the scale 1cm = 10 km, show the following information on a map

- Boomtown is 50 km due east of Alphaville
- Cooltown is 6 km south east of Boomtown
- Dogville is 20 km due east of Cooltown
- Eastborough is 30 km south west of Dogville
- Finishton is both due west of Eastborough and due south of Alphaville



5) A factory site has a main office and six workshops. Each workshop is the same distance from the office. Each workshop is exactly in between its neighbors. Show this information on a map



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6) Fill in the map below with the names of the states and capital cities of each one. Use a suitable scale to indicate distances, and supply some written information on relative locations (e.g. Perth is x km west of Sydney)

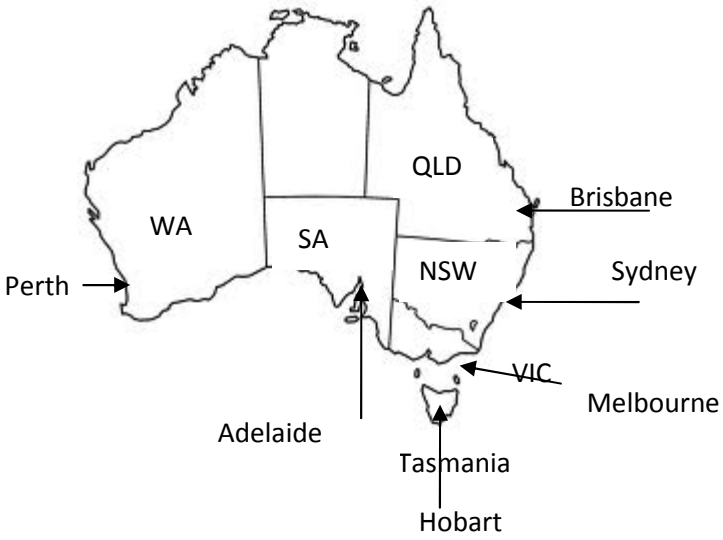
Scale 1cm ≅ 700 km

Adelaide is NW of Melbourne and Hobart

Brisbane is north of Sydney

Hobart and Melbourne are the closest capital cities to each other

There are more



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